Tutorial - 10 (Boosting)

sample	Feature1 (Age)	Feature2 (Income)	Feature3 (Credit Score)	Target (Loan Approved)
S1	25	40	650	Yes
S2	30	60	700	Yes
S3	45	35	600	No
S4	50	30	580	No
S5	28	55	720	Yes

Target Encoding:

Yes = +1

No = -1

① Assign initial weights to all 5 samples.

② Construct a decision stump using Feature1 (Age) to classify the samples

③ Identify misclassified samples and compute the weighted error rate.

④ Calculate the classifier weight.

⑤ update the sample weights for the next iteration.

1) Initial weights

N = no. of datapoints in dataset

$$w_i = \frac{1}{N} = \frac{1}{5} = 0.2$$

2) Decision stump :

Age < 40 -> Yes (+1)

≥ 40 -> No (-1)

Target	predicted	w
Yes	+1	0.2
Yes	+1	0.2
No	-1	0.2
No	-1	0.2
Yes	+1	0.2

3) Weighted error:

$$\varepsilon_i = \sum w_i \cdot I\,(y_i \neq h_i(x_i))$$

$$= 0.2 \times 0 + 0.2 \times 0 + \dots + 0.2 \times 0$$

$$= 0$$

As $y_i = h(x_i)$

$+1 \neq +1$

-> so it is not correct so $I = 0$

if it is correct $= I = 1$

both are equal

Not equal.

④ Classifier / Model Error.

$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$$

model a. is ∞ then it will stops working

$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1-0}{0}\right) = \infty$$

⑤ Update weights:

$$w_i^{new} \rightarrow w_i e^{-y_i \alpha_1 h_1(x_i)} \qquad \alpha = \infty$$

$$w_i^{new} \rightarrow w_i (e^{-\infty})$$

$$w_i^{New} = 0$$

⑥ prediction

model will stop training (x_i)

$H(x) = h_i(x_i)$

Target (y)	predict $h_1(x_1)$	(w_1)
yes (+1)	+1	0.2
yes (+1)	+1	0.2
NO (-1)	-1	0.2
NO (-1)	-1	0.2
NO (-1)	+1	0.2

① same

② "

③ weighted Error.

$$\varepsilon_1 = \sum w_i \cdot I\,(y_i \neq h_1(x))$$

$$= 0.2 \times 1 = 0.2$$

④ classifier / model error

$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$$

$$= \frac{1}{2} \ln\left(\frac{1-0.2}{0.2}\right)$$

$$= 0.693$$

⑤ updated weight:

w_i new $\rightarrow$ $w_i\, e^{-y_i \alpha_1 h_1(x_i)}$

$w_i = 0.2$
$y_i = 1$
$h_1 = 1$

$\Rightarrow$

w	wi new
0.2	0.1
0.2	0.1
0.2	0.1
0.2	0.1
0.2	0.4

weights are going to ↓ then it means the model is perfectly classified

* Normalized w.

$$= \frac{w_i}{\sum w_i} = \frac{0.2}{0.1+0.1+0.1+0.1+0.4}$$

$$= 0.8$$

$\Rightarrow$

WN
0.125
0.125
0.125
0.5

0.125 $\rightarrow$ all are decreased

as $\rightarrow$ this w is ↑ as 0.2 to 0.5

as the misclassification occured in the last classification 0.4 > 0.5 so the next w is prediction that.

③ Prediction

$$H(x) = \text{sign}\left[\sum_{t=1}^{T} \alpha_t h_t(x_i)\right]$$

$$= \text{sign}\left[\alpha_1 h_1(x_i) + \alpha_2 h_2(x_2) + \cdots\right]$$